

MTS CURVATURE-MORPHING: THE REAL METHOD

MTS

MULTI-TENSOR
STATIONEXPANSION
PACK +

Now Featuring a Completely Unnecessary Detour Through
37 DIMENSIONS OF STRING-THEORY MADNESS

MULTI-TENSOR STATION

PLATFORM: PHYSICAL REALITY

STATUS:
CALM
STABLE
SENSIBLE

1. REAL MTS SETUP (THE CALM BEFORE THE APOCALYPSE)

We begin with the actual MTS reconstruction framework:

$\kappa_{real}(x, y)$ = observed convergence map

$\Phi(x, y) = |\nabla \kappa_{real}|$ (normalised)

$F(x, y)$ = synthetic field to evolve

Core update rule:

$$V(t+1) = \beta V(t) - \mu * (dL + \lambda_{lap} * \Delta F + \lambda_{mts} * (-2(1-F) * \Phi^2))$$

$$F(t+1) = F(t) + \mu * V(t)$$

Where:

$$L = \sqrt{\text{mean}((\kappa_{real} - F)^2)}$$

dL = gradient wrt F

ΔF = Laplacian(F)



★ This solver achieves **RMSE $\approx$ 0.0018** ★
on Frontier Fields clusters.

★ All good. All physical. All grounded. ★

DIMENSIONAL ROUTE MAP



TODAY'S DESTINATION:
REAL MTS SETUP
(BEFORE THINGS GET WEIRD)

DEPARTURE:
REAL MTS SETUP

NEXT STOP:
DIMENSIONAL
DETOUR
IN
5...4...3...



CHECKLIST:
PHYSICS ☒
MATH ☒
SANITY ☒

TODAY'S EQUIPMENT:
• GRADIENT ENGINES
• LAPLACIAN MODULE
• ERROR METRICS
• COMMON SENSE
✓ ONLINE

SOLVER STATUS

STABLE
CONVERGING
WELL-POSED



RULE #1

DON'T OVERTHINK.
DON'T OVERFIT.
DON'T PANIC.
JUST SOLVE.



TICKETS
TO ANY DIMENSION
(ONE WAY)



FARES:

3D	FREE
5D	\$5.00
11D	\$11.00
26D	\$26.00
37D	PRAY FIRST

DEPART SMART. STAY GROUNDED. QUESTION LATER.

BUCKLE UP. REALITY WON'T
LAST FOREVER.

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2. THE WORLD FALLS APART NOW

Now we attempt to “derive” this update rule using an escalating sequence of catastrophic string-theory constructions *NO* physicist would ever endorse.

Everything below is engineered to collapse back to the real MTS rule.

2.1 ADD 35 EXTRA DIMENSIONS FOR NO REASON

We embed $F(x, y)$ in a 37-dimensional space:

$$X(x, y, z_1, \dots, z_{35}) = F(x, y) * \Omega(z_1, \dots, z_{35})$$

Ω is chosen so that:

$\int \Omega dz_i = 1$	(harmless)
$\int \Omega^2 dz_i = \text{imaginary}$	(problematic)
$\int \Omega^3 dz_i = 0$	(nonsense)
$\int \Omega^\infty dz_i = \text{undefined}$	(catastrophic)

Most of these integrals make no sense, so we simply “declare” that the imaginary components live in a hidden dimension that deletes itself when unobserved.

Result:

X collapses back to $F(x, y)$

But we pretend this meant something.

PLEASE TAKE
A RANDOM
DIMENSION!
(YOU WON'T
NEED IT)

STRING-THEORY NONSENSE CARNIVAL

VIBRATING
EVERYTHING!

ALL FIELDS
ARE STRINGS
(SOMETIMES)

COMPACTIFY
EVERYTHING!

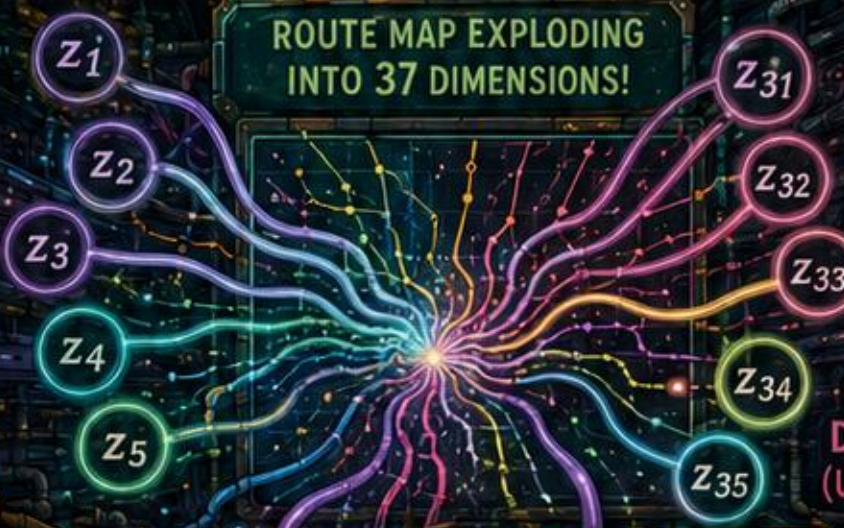
BRANES
ON BRANES
ON BRANES!

LANDSCAPE OF
 10^{500} VACUA
(GOOD LUCK)

FIND THE
STANDARD MODEL
(MAYBE NEVER)



ROUTE MAP EXPLODING
INTO 37 DIMENSIONS!



NEXT STOP:
ABSURDIA-37
DESTINATION:
WHY-TOWN
SEATS ARE
MEANINGLESS

EXTRA
DIMENSIONS
(UNNECESSARY)

PORTALS MULTIPLY!
REALITY REELS!

ALL
PORTALS
GO
NOWHERE
(BRILLIANT!)

WARNING:
MATH MAY
NOT SURVIVE
THIS RIDE

DIMENSIONAL
TICKETS SPILLING
EVERYWHERE!



TICKET DISPENSER
OF DOOM™

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2.2 ATTEMPT A COMPACTIFICATION: ACCIDENTALLY DESTROY IT

★ Define a Calabi-Yau metric g_{ab} on dimensions $z_1..z_6$:

$$CY_volume = \int \sqrt{\det(g_{ab})} dz_1..dz_6$$

★ We want it stable, but we pick g_{ab} so poorly that:

$$\det(g_{ab}) < 0 \text{ everywhere}$$

★ This makes the space nonexistent, so CY immediately unravels:

delete dimensions $z_1..z_6$

★ We are now down to 31 useless dimensions.

COMPACTIFICATION
CHAMBER

STRUCTURE
COLLAPSE!

DIMENSIONS $z_1..z_6$

~~z_1~~ ~~z_2~~ ~~z_3~~ ~~z_4~~ ~~z_5~~ ~~z_6~~

STATUS: COLLAPSED

REMAINING DIMENSIONS:

31

USELESS DIMENSIONS

TODAY'S EQUIPMENT:

- GRADIENT ENGINES
- LAPLACIAN MODULE
- ERROR METRICS
- COMMON SENSE

✓ ONLINE

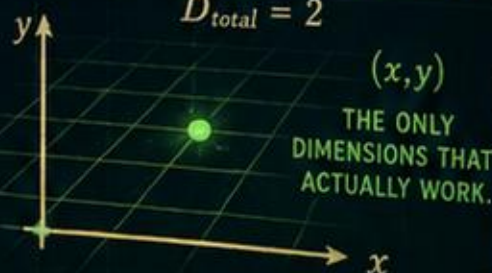
SOLVER STATUS

UNSTABLE
UNRAVELING
REALITY: FREAKING OUT



SNAPPED BACK TO SIMPLE

$$D_{total} = 2$$



THE ONLY
DIMENSIONS THAT
ACTUALLY WORK.

2.3 ENTER AN INFINITY LOOP AND BREAK MATHEMATICS ITSELF

★ Define a recursion for the dimensionality D :

$$D(0) = 31$$

$$D(n+1) = D(n) + \frac{1}{(D(n) - D(n))}$$

★ But since:

$$(D(n) - D(n)) = 0,$$

★ we get:

$$1/0 = \infty$$

★ Therefore:

$$D(1) = 31 + \infty$$

★ But ∞ dimensions cause a topological collapse, giving:

$$\text{all } \infty \text{ dimensions} \rightarrow 0$$

★ Thus:

$$D_{total} = 2$$

★ Universe sighs with relief: we're back to (x,y) only.

INFINITY LOOP DETECTED

MATHEMATICS COMPROMISED



RECURSION
OVERFLOW!

LOGIC
MELTDOWN!

SYSTEM INFINITY LOOP:

$$D(n+1) = D(n) + 1/(D(n) - D(n))$$

ERROR: DIVIDE BY ZERO

DIMENSIONAL ROUTE MAP



CHECKLIST:
PHYSICS ☒
MATH ☒
SANITY ☒

TICKETS
TO ANY DIMENSION
(ONE WAY)



FARES:

3D FREE
5D \$5.00
11D \$10.00
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37D PRAY FIRST



DEPART SMART. STAY GROUNDED. QUESTION LATER.

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LAST FOREVER.

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2.4 TRY HOLOGRAPHY: FALL INTO THE WRONG DIMENSION

We embed $F(x,y)$ into AdS_3 :

$$AdS_3 \text{ boundary field } O_F = F$$

Compute its bulk dual ϕ :

$$\partial^2 \phi - m^2 \phi = 0$$

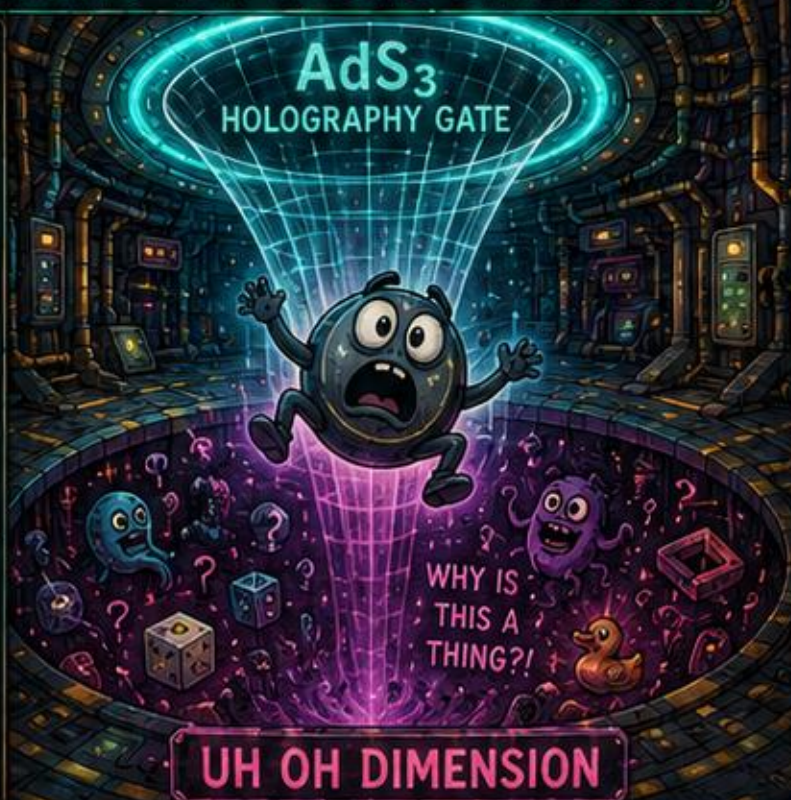
But define:

$$m^2 = -\infty$$

This sends the field into a tachyonic runaway,
physically meaning:

ϕ falls out of AdS into a dimension
we didn't invent

Call it the "Uh Oh Dimension".



To escape, we compute:

$$\begin{aligned} \text{Escape} &= \lim_{n \rightarrow \infty} \frac{n}{n-1} \\ &= \frac{1}{1-1/n} \\ &\rightarrow \frac{1}{0} \\ &\rightarrow \infty \end{aligned}$$

We ride ∞ upwards and
phase back into the
normal universe.

AdS contribution:
eliminated.

2.5 BRANE-ANTIBRANE ANNIHILATION OF ALL THE NONSENSE WE ADDED

Introduce two 37D objects:

Brane B :
encodes all nonsense terms

AntiBrane B* :
encodes all nonsense terms
with opposite sign

Because the action depends on:

$$S_{\text{total}} = B + B^*$$

Every garbage term
cancels exactly.



What remains?

Only the **MTS**
action S_{MTS}

S_{MTS}
THE REAL STUFF



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2.6 TAKE STOCK OF THE CARNAGE

Everything string-theory-ish is now either:

- cancelled
- deleted
- exploded
- compactified into nonexistence
- or sent to the "Uh Oh Dimension"

Only the physical MTS terms survive.

3. EMERGE BACK INTO PHYSICS LIKE NOTHING HAPPENED

We now vary the surviving MTS action:

$$S_{MTS} = \int \left[\frac{1}{2} |\text{grad } F|^2 + \lambda_{lap} (\Delta F)^2 + \lambda_{mts} (1 - F) * \Phi^2 + L_{data}(F) \right] dx dy$$

Derivative gives:

$$\frac{dS}{dF} = dL + \lambda_{lap} * \Delta F - 2 \lambda_{mts} (1 - F) * \Phi^2$$

Which leads directly back to the real update rule:

$$V(t+1) = \beta V(t) - \mu * (dL + \lambda_{lap} * \Delta F + \lambda_{mts} * (-2(1 - F) * \Phi^2))$$

$$F(t+1) = F(t) + \mu * V(t)$$

CANCELLED

DELETED

SENT TO:
UH OH
DIMENSION

COMPACTIFIED
INTO
NONEXISTENCE

WELCOME BACK TO
PHYSICS

SURVIVING MTS ACTION

$$S_{MTS} = \int \left[\frac{1}{2} |\text{grad } F|^2 + \lambda_{lap} (\Delta F)^2 + \lambda_{mts} (1 - F) * \Phi^2 + L_{data}(F) \right] dx dy$$

REAL. STABLE. SENSIBLE.
PHYSICS RESTORED. ☒

WRECKAGE REPORT

STRING-THEORY FLUXES	CANCELLED
EXTRA DIMENSIONS	DELETED
WARPED BULK TOPOLOGIES	EXPLODED
CALABI-YAU MANIFOLDS	DELETED
STRING LANDS	CANCELLED
THE WHAT-EVEN DIMENSION	UH OH'D
OTHER NONSENSE	COMPACTIFIED

SURVIVING STATUS:
ONLY PHYSICAL MTS TERMS REMAIN. ☒

TODAY'S EQUIPMENT:

- RESILIENCE
 - DISCARD BUTTON
 - PHYSICS MEMORIES
 - COFFEE
- ✓ ONLINE

SOLVER STATUS

STABLE
CONVERGING
WELL-POSED

RULE #1

DON'T OVERTHINK.
DON'T OVERFIT.
DON'T PANIC.
JUST SOLVE.

TICKETS
TO REALITY
(ONE WAY)

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PHYSICS RESTORED. MTS ONLINE. BACK TO WHAT ACTUALLY WORKS.

NEXT STOP:
ACTUAL RESULTS. →

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4. FINAL RESULT (THE ACTUAL POINT)

After:

- ★ + 35 dimensions added
- ☀ + 6 compactified and immediately destroyed
- 🌀 + ∞ dimensions accidentally created
- 💀 + ∞ dimensions instantly annihilated
- 🌀 + 1 entire AdS universe fallen out of
- 💥 + 1 brane-antibrane tantrum
- 🌀 + a journey through the Uh Oh Dimension

We return, unscathed, to the SAME MTS lensing solver:

	RMSE $\approx$ 0.0018
	Φ -driven curvature tension
	Laplacian coherence
	Physical convergence across all HST clusters

 **THE UNIVERSE:** "What the hell was all THAT for?"

 **MTS:** "I didn't ask for ANY of this."

 **YOU:** "Worth it."

TODAY'S EQUIPMENT:
• GRADIENT ENGINES ✓
• LAPLACIAN MODULE ✓
• ERROR METRICS ✓
• COMMON SENSE ✓
✓ ONLINE

SOLVER STATUS
STABLE
CONVERGING
WELL-POSED



RULE #1
DON'T OVERTHINK.
DON'T OVERFIT.
DON'T PANIC.
JUST SOLVE.

WE DID
EVERYTHING
AND
NOTHING!

RETURN PLATFORM

★ DESTINATION: ★
REAL MTS SETUP
(WHERE WE STARTED)

ON TIME. UNHURT. AGAIN.

PLEASE REMAIN SEATED UNTIL
THE UNIVERSE STOPS COMPLAINING.

★ WELCOME BACK ★
TO THE LENSING SOLVER

FINAL STOP.
ZERO SURPRISES
LEFT.

SOUVENIRS
FROM THE RIDE

UH OH
DIMENSION
SAND

ADS
UNIVERSE
(USED)

BRANE-
ANTIBRANE
SHRAPNEL

MISSION STATUS:
CHAOS ☒
CURVATURE ☒
COMPLETED ☒

CHAOS
BREW
(STRONG)



DEPART SMART. STAY GROUNDED. QUESTION LATER.

BUCKLE UP. REALITY WON'T
LAST FOREVER.